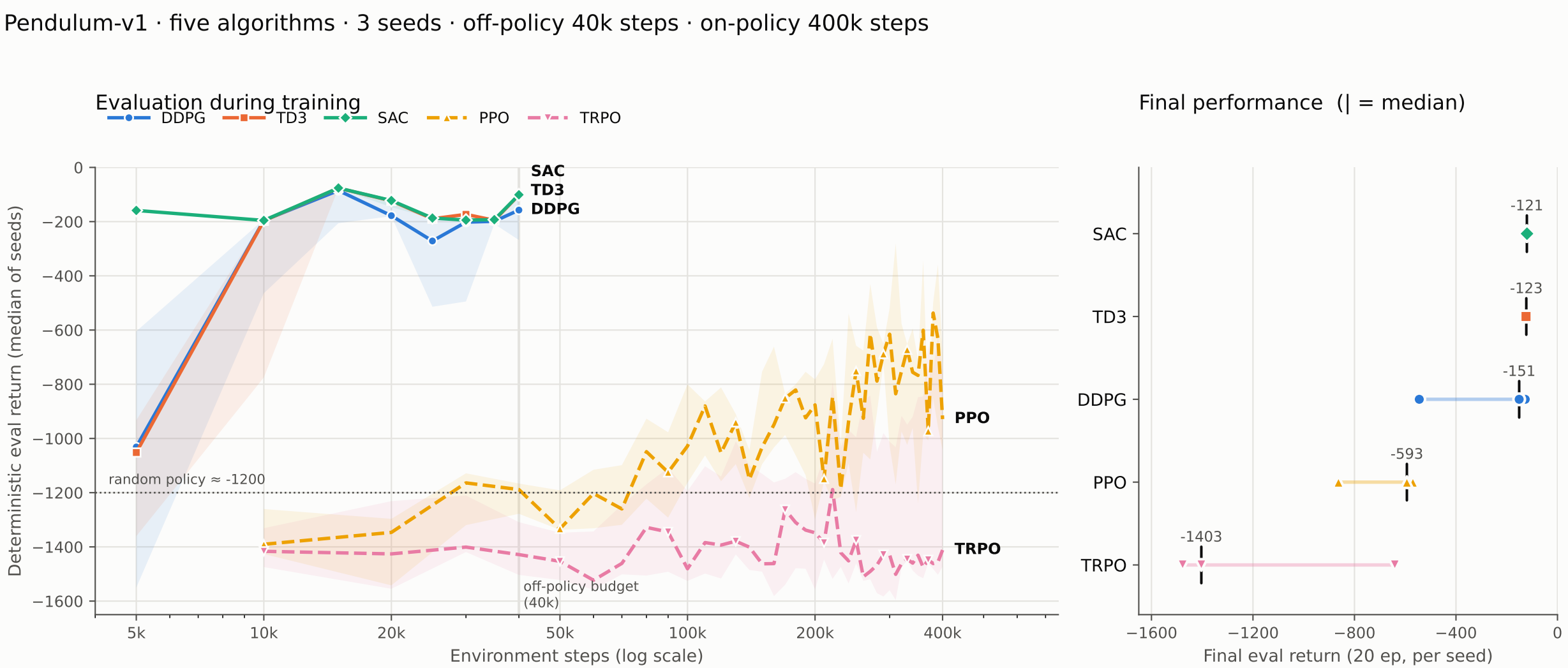
Pendulum-v1 · five algorithms · 3 seeds · off-policy 40k steps · on-policy 400k steps
Evaluation during training
DDPG
TD3
SAC
PPO
TRPO
Deterministic eval return (median of seeds)
0
−200
−400
−600
−800
−1000
−1200
−1400
−1600
SAC
TD3
DDPG
PPO
TRPO
random policy ≈ -1200
off-policy budget
(40k)
5k
10k
20k
50k
100k
200k
400k
Environment steps (log scale)
Final performance (| = median)
SAC
TD3
DDPG
PPO
TRPO
-121
-123
-151
-593
-1403
−1600
−1200
−800
−400
0
Final eval return (20 ep, per seed)